

Solutions to Year 2 Quantum Mechanics Unassessed Problem Sheet 3:

1. Given:

$$E = \frac{\pi^2 \hbar^2}{2ma^2} (n_x^2 + n_y^2 + n_z^2),$$

- a) the lowest possible value for E is $\frac{3\pi^2 \hbar^2}{2ma^2}$, with a degeneracy of 1 (i.e. no degeneracy);
- b) the second lowest value for E is $\frac{6\pi^2 \hbar^2}{2ma^2}$, with a degeneracy of 3 (one of the indices = 2, other 2 = 1; 3 possible choices). The three states could be distinguished from the probability functions, since whichever coordinate has $n = 2$ will have a node in the centre of the well in that direction.

c)

n^2	3	6	9	11	12
n_x	1	2 1 1	2 2 1	3 1 1	2
n_y	1	1 2 1	2 1 2	1 3 1	2
n_z	1	1 1 2	1 2 2	1 1 3	2
degeneracy	1	3	3	3	1

For a degeneracy of 6 just try all 3 n 's different, e.g. $3^2 + 2^2 + 1^2 = 14$, for which there are 6 permutations.

2. A particle in an infinite square well extending from $x = 0$ to $x = L$ has the wavefunction:

$$\Psi(x, t) = A \left(2 \sin\left(\frac{\pi x}{L}\right) e^{-iE_1 t/\hbar} + \sin\left(\frac{2\pi x}{L}\right) e^{-iE_2 t/\hbar} \right),$$

where $A = \sqrt{2/5L}$.

The probability density $P(x, t) = \Psi^* \Psi$ is then:

$$\begin{aligned} P(x, t) &= \frac{2}{5L} \left(2 \sin\left(\frac{\pi x}{L}\right) e^{iE_1 t/\hbar} + \sin\left(\frac{2\pi x}{L}\right) e^{iE_2 t/\hbar} \right) \left(2 \sin\left(\frac{\pi x}{L}\right) e^{-iE_1 t/\hbar} + \sin\left(\frac{2\pi x}{L}\right) e^{-iE_2 t/\hbar} \right) \\ &= \frac{2}{5L} \left[4 \sin^2\left(\frac{\pi x}{L}\right) + \sin^2\left(\frac{2\pi x}{L}\right) + 2 \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{2\pi x}{L}\right) \left(e^{i(E_1 - E_2)t/\hbar} + e^{i(E_2 - E_1)t/\hbar} \right) \right] \\ &= \frac{2}{5L} \left[4 \sin^2\left(\frac{\pi x}{L}\right) + \sin^2\left(\frac{2\pi x}{L}\right) + 4 \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{2\pi x}{L}\right) \cos\left(\frac{(E_2 - E_1)t}{\hbar}\right) \right]. \end{aligned}$$

Therefore the probability density oscillates with an angular frequency $\omega = \frac{E_2 - E_1}{\hbar}$.

However, when $x = \frac{L}{2}$ then $\sin\left(\frac{2\pi x}{L}\right) = 0$ and so $P(x, t) = \frac{8}{5L} = \text{constant}$.

$$\begin{aligned}
 \langle p_x \rangle &= \int_0^L \Psi^* \left(-i\hbar \frac{\partial}{\partial x} \right) \Psi dx \\
 &= \frac{-2i\hbar}{5L} \int_0^L \left(2 \sin \frac{\pi x}{L} e^{iE_1 t/\hbar} + \sin \frac{2\pi x}{L} e^{iE_2 t/\hbar} \right) \left(\frac{2\pi}{L} \cos \frac{\pi x}{L} e^{-iE_1 t/\hbar} + \frac{2\pi}{L} \cos \frac{2\pi x}{L} e^{-iE_2 t/\hbar} \right) dx \\
 &= \frac{-2i\hbar}{5L} \int_0^L \frac{4\pi}{L} \cos \frac{\pi x}{L} \sin \frac{\pi x}{L} + \frac{2\pi}{L} \cos \frac{2\pi x}{L} \sin \frac{2\pi x}{L} + \frac{2\pi}{L} \sin \frac{2\pi x}{L} \cos \frac{\pi x}{L} e^{i(E_2 - E_1)t/\hbar} \\
 &\quad + \frac{4\pi}{L} \sin \frac{\pi x}{L} \cos \frac{2\pi x}{L} e^{i(E_1 - E_2)t/\hbar} dx.
 \end{aligned}$$

Now with $\omega = (E_2 - E_1)/\hbar$ and using the integrals given and trigonometry formulae, we obtain

$$\begin{aligned}
 \langle p_x \rangle &= \frac{-2i\hbar}{5L} \int_0^L \left[\frac{2\pi}{L} \left(\sin \frac{3\pi x}{L} - \sin \frac{\pi x}{L} \right) e^{-i\omega t} + \frac{\pi}{L} \left(\sin \frac{3\pi x}{L} + \sin \frac{\pi x}{L} \right) e^{i\omega t} \right] dx \\
 &= \frac{-2i\hbar}{5L} \left[\frac{2\pi}{L} \left(-\frac{L}{3\pi} \cos \frac{3\pi x}{L} + \frac{L}{\pi} \cos \frac{\pi x}{L} \right) e^{-i\omega t} + \frac{\pi}{L} \left(-\frac{L}{3\pi} \cos \frac{3\pi x}{L} - \frac{L}{\pi} \cos \frac{\pi x}{L} \right) e^{i\omega t} \right]_0^L \\
 &= \frac{-2i\hbar}{5L} \left[\frac{2\pi}{L} \left(\frac{2L}{3\pi} - \frac{2L}{\pi} \right) e^{-i\omega t} + \frac{\pi}{L} \left(\frac{2L}{3\pi} + \frac{2L}{\pi} \right) e^{i\omega t} \right] \\
 &= \frac{-2i\hbar}{5L} \left[\frac{-8}{3} e^{-i\omega t} + \frac{8}{3} e^{i\omega t} \right] \\
 &= \frac{-2i\hbar}{5L} \left[\frac{8}{3} 2i \sin \omega t \right] \\
 &= \frac{32\hbar}{15L} \sin(\omega t).
 \end{aligned}$$

The sketch is just a sine wave, amplitude $\frac{32\hbar}{15L}$.

The sign of $\langle p_x \rangle$ changes every half-period, indicating that the probability density is moving back and forth in the well with an angular frequency of ω .

3. Mean lifetime $\tau = 1/\gamma = 10^{-8}$ s. Energy of photons $= hc/\lambda = 3.3 \times 10^{-19}$ J.

Uncertainty relation gives $\Delta E \tau \geq \hbar/2$.

$$\therefore \Delta E \approx \frac{\hbar}{2\tau} = 5.3 \times 10^{-27} \text{ J}.$$

$$E = \frac{hc}{\lambda} \Rightarrow \lambda = \frac{hc}{E}$$

$$\therefore \Delta\lambda = \frac{d\lambda}{dE} \Delta E$$

$$= -\frac{hc}{E^2} \Delta E$$

$$|\Delta\lambda| = 9.5 \times 10^{-15} \text{ m}$$

